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What Stair Wear Can Remember

Identifiable Patterns and Defensible Historical Bounds

A worn stair records accumulated contact, but it does not preserve a unique history. Our non-destructive framework separates measurable surface loss and pathways from conditional estimates of traffic, direction and age. A scalar volume budget and calibrated spatial mixture identify the claims a surface can support.

We prove three limits. Increasing traversals and inversely reducing wear per traversal preserves the surface. Reordering the same foot contacts preserves additive wear, so lateral lanes cannot establish simultaneous side-by-side use or opposing flow. Compressing traffic into a shorter interval also preserves wear. Material calibration and chronological evidence are therefore needed to estimate daily activity or age.

For direction, ascent and descent kernels must remain linearly independent after reference-surface and environmental nuisance patterns are removed. Even then, their coefficients measure wear contribution rather than person counts. A labeled synthetic $1.20\,\mathrm{m} \times 0.30\,\mathrm{m}$ tread with true ascent wear share 0.700 yielded 0.701 under independent depth noise. A shifted kernel produced a detectable residual, while cross-mixed directional kernels biased the share to 0.640 with zero residual. We derive an exact bounded-material feasibility condition and compute count-share intervals. With a reference directional wear-coefficient ratio allowed up to two, balanced counts reproduce the apparent 70:30 field using approximately $\pm 7.69\%$ local susceptibility variation. In a specified 0.30 mm bounded-error scenario, the majority guarantee disappears at about 3.54%. These thresholds are synthetic discrimination results, not measured material limits.

Illustrative loss of 580–$620\,\mathrm{cm}^3$, stipulated wear of 0.02–$0.08\,\mathrm{mm}^3$ per traversal and independent age of 300–500 years permit approximately 40–283 traversals per calendar day. These are scenario bounds, not estimates for the photograph. The same 15 million traversals leave identical wear over 400 years at 103 per day or 40 years at 1,027 per day. A low-wear tread locates a repair candidate; harder material or reduced contact remains an alternative.

Our practical survey needs two people, a phone camera, a tape, a straightedge with a gap scale, a flashlight and a hand lens. A 24-point design spanning front and rear retained much of a 3,321-point scan's precision under the tested correlated errors; central-only measurements lost direction entirely. Protected references, joints and visible material features target the more consequential ambiguities. Historical consistency is the existence of a plausible parameter set, and provenance requires candidate-material comparison; wear alone cannot name a quarry or tree.

Keywords: cumulative abrasion; inverse problem; identifiability; material confounding; archaeological measurement.

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1 The archaeological questions and the evidence available

A stair surface provides a spatial record of loss. The questions ask for a temporal and social history: use frequency, preferred direction, simultaneous occupancy, construction age, repairs and material origin. These are related inverse problems with different information requirements. We use one wear model to determine which conclusions follow from shape, which require independent history, and which the surface cannot determine.

The statement supplies an example photograph, not a scaled survey or a material dataset. We do not extract millimetric depths, ages or traffic counts from that photograph. Numerical observations in this paper are either labeled synthetic tests or stipulated illustrations. They demonstrate what the proposed inference would do after an archaeologist supplies measurements. Table 1 connects the original questions to our answers.

Table 1. Questions and the evidence required to answer them.

Question	Defensible answer	Location
Frequency and typical day	Traversal bounds after wear calibration and age; people require trip frequency	Sections 3, 7
Favored direction	Conditional spatial mixture and a material/kernel sensitivity check	Section 4
Simultaneous use	Spatial paths, geometric feasibility; concurrency remains ambiguous	Section 5
Measurements	Low-cost repeated profiles and protected controls	Section 6
History consistency and age	Feasible parameter set and conditional age interval	Section 7
Repairs	Cross-tread anomalies plus joints/material evidence	Section 8
Material source	Candidate compatibility; independent visible or compositional features	Section 9

2 A cumulative wear model with explicit observation limits

2.1 Surface geometry, units and the forward mechanism

For tread i, let x be lateral position and y distance inward from the nosing. Let $z_i^0(x, y)$ denote its original surface and $z_i(x, y)$ its present surface, in millimetres. The loss is $h_i = z_i^0 - z_i$. Reconstructing z_i^0 is part of the measurement problem; fitting a plane through the worn center would erase the loss we want to measure. Protected edges, sheltered patches or an intact geometrically matched tread provide candidate references. We retain more than one reference when construction camber or slope is uncertain.

In the Archard-inspired baseline, a contact loses volume

$$q = \frac{kF\ell}{H}, \tag{1}$$

where F is normal force in newtons, ℓ is effective sliding distance in millimetres, H is hardness in $\mathrm{N/mm^2}$ and k is dimensionless. Thus q has units $\mathrm{mm^3}$. This phenomenological relation motivates a budget, rather than specifying a universal stone constant. Tribometer studies calibrate wear coefficients from the relevant contact system and limit their transfer range [1]. Shoe material, grit, moisture, grain orientation and gait enter k and ℓ; hardness alone cannot supply q. Peak plantar pressure is likewise insufficient because stationary loading need not cause sliding abrasion.

Let $K_{i,d}(x, y)$ be expected loss shape for one traversal in direction $d \in \{u, v\}$, normalized so that $\int_{A_i} K_{i,d}\, dA = 1$. Its units are $\mathrm{mm^{-2}}$. Let $n_d(t)$ be traversals per year and $q_{i,d}(t)$ loss per traversal per tread. Between repairs,

$$h_i(x, y) = \sum_d \int_0^T q_{i,d}(t) n_d(t) K_{i,d}(x, y, t)\, dt + e_i(x, y), \tag{2}$$

where e_i includes non-traffic loss, such as erosion. The volume is $V_i = \int_{A_i} h_i\, dA$. Surface chips, deep rot, sanding and replacement require separate terms or epochs, not an artificially large abrasion coefficient.

With constant kernels and coefficients, define $N_d = \int_0^T n_d(t)dt$ and $a_{i,d} = q_{i,d}N_d$. Equation (2) reduces to

$$h_i = a_{i,u}K_{i,u} + a_{i,v}K_{i,v} + e_i. \tag{3}$$

Here $a_{i,d}$ is a volume, not a number of people. Integrating across m treads counts contact loss on m treads; it does not imply m times as many stairwell traversals.

2.2 Working assumptions and alternatives

Local loss is approximately additive when the depression is small enough that contact patterns remain stable. We test changed kernels, a saturating exposure law and retention epochs. Strong geometry adaptation, fatigue fractures or nonlinear crowd loading require new calibration. A result derived under additive abrasion remains a conditional interpretation.

A common traffic pattern across successive treads is plausible for a flight without branching entrances. We check it by comparing normalized fields, not by assuming all treads have the same loss. Material and foot loading can differ with tread geometry. Landing treads, winders and bottlenecks are analyzed separately. Reference offsets and tilts enter the observation model as nuisance effects. Spatial material variation remains a structural alternative because it can imitate a change in direction.

Table 2 separates measured quantities from inferred ones. The distinction prevents an assumed coefficient or construction date from acquiring the status of an observation.

Table 2. Quantities, their roles and how they are obtained.

Quantity	Meaning and unit	Evidence status
h_i, V_i	Loss depth (mm), integrated volume (mm^3)	Measured relative to candidate original surface
$K_{i,d}$	Normalized spatial loss per traversal (mm^{-2})	Calibration assumption; compare geometry/gait alternatives
$q_{i,d}$	Effective volume per traversal (mm^3)	Matching contact calibration or documented range
N_d, T	Traversals, retained exposure years	Conditional inverse quantities; chronology supplied independently
e_i, r_{ij}	Environmental loss (mm), retained fraction of epoch j	Protected controls and evidence of intervention

3 Use frequency: from loss volume to conditional traffic

3.1 What the volume budget identifies

After estimating non-traffic loss, let $V^{\mathrm{tr}} = V - V^e$ be traffic-attributable volume on a representative tread. If the effective per-traversal coefficient is $\bar{q}$, then

$$N = \frac{V^{\mathrm{tr}}}{\bar{q}}, \qquad \bar{n} = \frac{N}{365.25T}, \tag{4}$$

where $\bar{n}$ is mean traversals per calendar day. If the stairs were open for a fraction f of days, the mean per active day is $\bar{n}/f$. A traversal is one movement through the flight in one direction. A resident making four daily traversals contributes four units, so $\bar{n}$ is not a census. Population requires an independent estimate of mean traversals per person.

Proposition 3.1 (Scale ambiguity)**.** *In Equation* (3)*, with unknown positive q_d and N_d, the surface determines their products at most. For every $c_d > 0$, the substitutions $N'_d = c_d N_d$ and $q'_d = q_d/c_d$ preserve the entire field.*

Proof. Each term becomes $q'_d N'_d K_d = q_d N_d K_d$. Every point and the integrated volume remain identical. Additional measurements of the same field cannot separate the two factors. □

The implication is practical: a dense depth scan can improve V but cannot independently calibrate $\bar{q}$. Recorded traffic on a comparable stair, an existing matched-material abrasion study, or repeated survey with independently counted current traversals can constrain q. A contemporary survey must still assess transfer to historic shoes, grit and environmental conditions. Repeated depth measurement is informative only when the change exceeds registration uncertainty; a short observation period may reveal no resolvable abrasion.

3.2 A transparent numerical illustration

Suppose measured traffic loss is bounded by 580–620 cm^3, an independent history gives $T \in [300, 500]$ years, and matching-contact evidence were to justify $\bar{q} \in [0.02, 0.08]$ mm^3 per traversal. The latter range is stipulated for this illustration, not measured or claimed typical for stone. Monotonicity in Equation (4) gives

$$N \in [7.25, 31.0] \times 10^6, \qquad \bar{n} \in [39.69, 282.92] \text{ traversals/day}. \tag{5}$$

The daily endpoints are rounded outward. With $V = 600\,\mathrm{cm}^3$, $\bar{q} = 0.04\,\mathrm{mm}^3$ and $T = 400$ years, the central scenario is 15 million traversals and 102.67 per day. These conditional results answer how to infer frequency without assigning an unsupported archaeological number.

All three uncertainties matter, but they do not matter equally. A fourfold coefficient range dominates a 6.9% volume range; increasing survey density further would not materially narrow Equation (5). For a small perturbation,

$$\frac{d\bar{n}}{\bar{n}} = \frac{dV^{\mathrm{tr}}}{V^{\mathrm{tr}}} - \frac{d\bar{q}}{\bar{q}} - \frac{dT}{T}. \tag{6}$$

If probabilistic uncertainties and their covariance are supported, this derivative propagates them. Otherwise we use scenario bounds rather than an invented confidence level. Common reference error must not be treated as independent error at thousands of scan pixels.

4 Preferred direction: a conditional spatial inverse problem

4.1 The rank criterion and the meaning of a mixture

Direction can be inferred only if the two directions have distinct, applicable loss kernels. Biomechanical studies show that plantar loading depends on activity and foot region [2]. They do not establish a universal rule that a front depression always means ascent. The signed association must come from a comparable mock staircase or an independently documented use pattern. Pressure, foot placement and slipping should be observed together; a temporary pressure mat or washable contact medium on a replica can support placement calibration without touching the monument.

Write measured depths as $\mathbf{w} = X\mathbf{a} + B\mathbf{c} + \boldsymbol{\epsilon}$. Columns of X contain sampled K_u, K_v; B contains plausible nuisance fields, such as a constant and reference tilt in x and y. For unequal or correlated measurement errors, first whiten by the measured covariance. In whitened coordinates, let $P = I - BB^{+}$ be the orthogonal projector outside the nuisance span.

Proposition 4.1 (Identifiable directional contributions)**.** *For unrestricted nuisance coefficients and positive directional contributions in the interior, the two contributions are identifiable from noiseless depths if and only if PX has rank two. When rank is two, the least-squares estimate satisfies*

$$\|\widehat{\mathbf{a}} - \mathbf{a}\|_2 \le \frac{\|P\boldsymbol{\epsilon}\|_2}{s_{\min}(PX)}. \tag{7}$$

Proof. Two solutions differ by $X\Delta\mathbf{a} + B\Delta\mathbf{c} = 0$. Projection yields $PX\Delta\mathbf{a} = 0$. Rank two forces $\Delta\mathbf{a} = 0$; rank deficiency admits a nonzero difference, which can be scaled small enough to retain positivity, and its component in the nuisance span is canceled by $\Delta\mathbf{c}$. For the error bound, the projected least-squares error is $(PX)^{+}P\boldsymbol{\epsilon}$, whose operator norm is $1/s_{\min}(PX)$. □

If the kernels are identical or differ only by a nuisance field, direction is unobservable. If their separation is small, noise or calibration error is amplified. One central cross-section with no front-to-back information may discard the very feature that distinguishes directions. The singular-value criterion therefore also guides where to measure.

The wear share is $\theta_w = a_u/(a_u + a_v)$. The traversal share is

$$\theta_N = \frac{\theta_w q_v}{\theta_w q_v + (1 - \theta_w) q_u}. \tag{8}$$

For $\theta_w = 0.70$ and $q_u/q_v \in [0.5, 2]$, the implied ascent traversal share is 0.538–0.824. Equal coefficients would yield 0.70, but that equality must be justified. A claim of favored ascent requires the lower feasible count share to exceed 0.5, with kernels, material variation and measurement error included.

4.2 Synthetic tests of precision, calibration and confounding

We generated a 1,200 × 300 mm tread on an 81 × 41 grid. Lateral pathways lie at $x = \pm 270$ mm; each has two alternating foot offsets of ±55 mm, broadened by Gaussian standard deviation 65 mm. The assigned ascent and descent kernels have Gaussian y centers at 100 and 200 mm with standard deviation 30 mm, normalized to unit integrated area. These are designed kernels, not empirical ancient foot maps. Their direction labels are hypothetical calibration labels.

Total traffic volume is 600 cm^3 and $\theta_w = 0.70$. Independent depth noise has standard deviation 0.15 mm; the example also includes a 0.20 mm common offset and tilts, estimated as nuisance coefficients. Figure 1 displays the directional information used by the inverse model.

Table 3. Synthetic direction tests under the same fitting model.

Generated surface	Estimated θ_w	Residual RMS (mm)
Nominal kernels, noisy surface	0.7010	0.1473
Kernels shifted 15 mm, width 34 mm	0.6563	0.7592
Directional kernels cross-mixed by 15%	0.6400	$< 10^{-12}$
Exposure saturation with 8 mm limit	0.6421	0.2372
Balanced traffic, varying material susceptibility	0.7000	$< 10^{-12}$

The nominal estimate in Table 3 is numerically accurate. In 400 synthetic repetitions with independent noise, the empirical 2.5–97.5 percentile range was $[0.6989, 0.7011]$. This describes that noise model, not archaeological uncertainty. A shifted, broader kernel changed the inferred

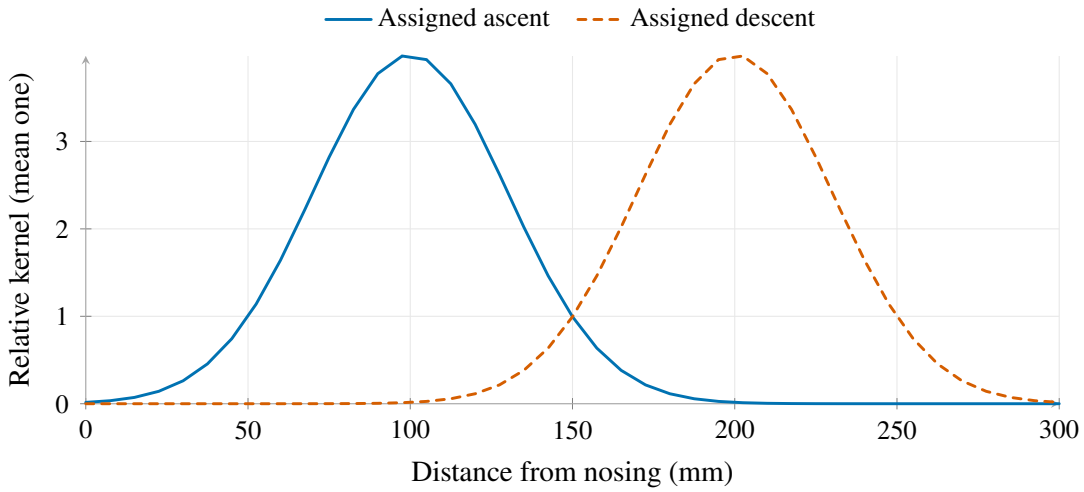


Figure 1. Distinct front-to-back kernels in the synthetic calibration.

share and left a residual five times the noise scale, so a residual check rejects reuse of the nominal calibration.

Residuals do not detect every kernel error. Define actual kernels $K'_u = 0.85K_u + 0.15K_v$ and $K'_v = 0.15K_u + 0.85K_v$. With $q_u = q_v$ and true ascent count share 0.70, fitting the nominal kernels gives $0.15 + 0.70(0.70) = 0.64$ exactly. Both kernels still integrate to one; only their directional associations changed. A separate stylized alternative sets $h = 8[1 - \exp(-h_{\text{linear}}/8)]$ in millimetres. It reduces the synthetic volume from 600 to 443.34 cm^3 and biases the fitted share to 0.6421. This is an exposure-dependent saturation challenge, not a validated stone law. The nominal linear peak was 11.96 mm, so checking geometry-induced contact changes is consequential. These alternatives require calibration evidence beyond a good fit.

A more dangerous alternative passes the residual check exactly. Let

$$M(x, y) = \frac{0.7K_u(x, y) + 0.3K_v(x, y)}{0.5K_u(x, y) + 0.5K_v(x, y)}. \tag{9}$$

Both kernels are positive on the finite grid, so M is defined. Equal traffic with effective local susceptibility M gives exactly the 70:30 surface. In this example, $M \in [0.60, 1.40]$; the maximum field discrepancy was 1.8×10^{-15} mm. The physical point is not that this particular variation is known to occur. It is that a ±40% allowed spatial susceptibility variation defeats a direction claim despite an essentially perfect fit. Protected and worn material patches, grain direction, coatings and moisture paths should therefore be checked before interpreting asymmetry.

An independent implementation solved the normal equations rather than using the author's pseudoinverse and recovered $\theta_w = 0.7010475$. A Schur-complement calculation confirmed projected singular values 120.283 and 111.502 for the mean-one computational kernels. Independent Simpson integration differed from their trapezoidal normalization by 0.0027%. These tests verify calculation; Equation (9) tests the more consequential structural interpretation.

4.3 How much material uncertainty defeats a direction claim?

The preceding counterexample can be strengthened into an interpretable feasible interval. Let $\kappa_d = AK_d$ be dimensionless mean-one kernels and let q_d^0 denote loss coefficients for a homogeneous reference material. For a direction-independent local susceptibility multiplier M, write the corrected depth field as

$$w = sMf_\phi + \epsilon, \qquad f_\phi = \phi\kappa_u + (1-\phi)\kappa_v, \quad 1-\delta \le M \le 1+\delta, \quad |\epsilon| \le e. \tag{10}$$

Here $s = (q_u^0 N_u + q_v^0 N_v)/A > 0$ has units mm and $\phi = q_u^0 N_u/(q_u^0 N_u + q_v^0 N_v)$. The parameter ϕ is a reference-material mixture, not the actual integrated wear share after M varies. Its conversion to counts uses Equation (8) with ϕ and q_d^0; using actual post-M coefficients there would count material uncertainty twice. The bounds δ, e must come from material checks and depth uncertainty. Neither is inferred from goodness of fit. The corrected field assumes nuisance removal has been justified; uncertainty remaining in that correction belongs in e.

Proposition 4.2 (Exact material-envelope criterion)**.** *For positive fixed kernels and a positive noiseless depth field w, a candidate ϕ is feasible with some $s > 0$ and $M \in [1-\delta, 1+\delta]$, $0 \le \delta < 1$, if and only if*

$$\frac{r_{\max}}{r_{\min}} \le \frac{1+\delta}{1-\delta}, \qquad r = \frac{w}{f_\phi}. \tag{11}$$

The smallest feasible variation is $(r_{\max} - r_{\min})/(r_{\max} + r_{\min})$.

Proof. At each location, $M = r/s$. Thus a common amplitude exists precisely when $r_{\max}/(1+\delta) \le s \le r_{\min}/(1-\delta)$. This is Equation (11). Solving it for δ gives the stated minimum. At that minimum, choose $s = (r_{\max} + r_{\min})/2$; the resulting multiplier reaches the two allowed endpoints. □

For a continuous field the extrema are over the tread; for a sampled field the condition applies at the observed points. It does not certify behavior between unmeasured locations. Allowing M to vary freely pointwise is conservative; geological smoothness or independent patch measurements can tighten the feasible set.

With bounded depth error, define $l = \max(0, w-e)/(1+\delta)$ and $u = (w+e)/(1-\delta)$. A candidate is feasible if

$$\max_j \frac{l_j}{f_{\phi,j}} \le \min_j \frac{u_j}{f_{\phi,j}}. \tag{12}$$

This follows by intersecting the permitted amplitude intervals at every point. Equivalently, setting $t = 1/s$ gives linear inequalities $l_j t \le f_{\phi,j} \le u_j t$ in two variables (ϕ, t), with $0 \le \phi \le 1$ and $t > 0$. Minimizing and maximizing ϕ provides continuous-parameter bounds without an arbitrary grid of candidate mixtures. We solved this finite-pixel problem and independently checked the amplitude-interval endpoints, primal constraints and stored dual certificates. The largest endpoint discrepancy was 1.1×10^{-8}; the largest normalized primal violation was 8.6×10^{-10}. These are floating-point verification tolerances, distinct from the exact proposition.

For the synthetic 70:30 field and $q_u^0/q_v^0 \in [0.5, 2]$, favored ascent requires $\phi > 2/3$. Proposition 4.2 gives a critical $\delta \approx 0.0769$. At that bound, $\phi = 2/3$, $q_u^0/q_v^0 = 2$ and hence exactly balanced

counts reproduce the field using a multiplier ranging approximately from 0.9231 to 1.0769 and a changed overall amplitude. The discretized threshold remained unchanged when the front-to-back grid was refined from 41 to 81 and 161 points; the nearly separated-kernel limit is $1/13$. These checks concern the specified kernels, not a universal archaeological threshold.

Figure 2 shows the lowest feasible ascent count share as δ increases. The second scenario measures $w = h + 0.15 + 0.15\sin(x/100)\cos(y/50)$ in millimetres and declares $e = 0.30\,\text{mm}$; its actual errors lie within that bound. This deterministic challenge is not a Gaussian confidence interval. In it, the majority threshold falls to about 3.54%. At $\delta = 5\%$, the clean lower count share is 0.514, but the bounded-error lower share is 0.493, so favored ascent is no longer established. A field report should supply its own defensible bounds and report an unresolved direction whenever the interval contains $1/2$.

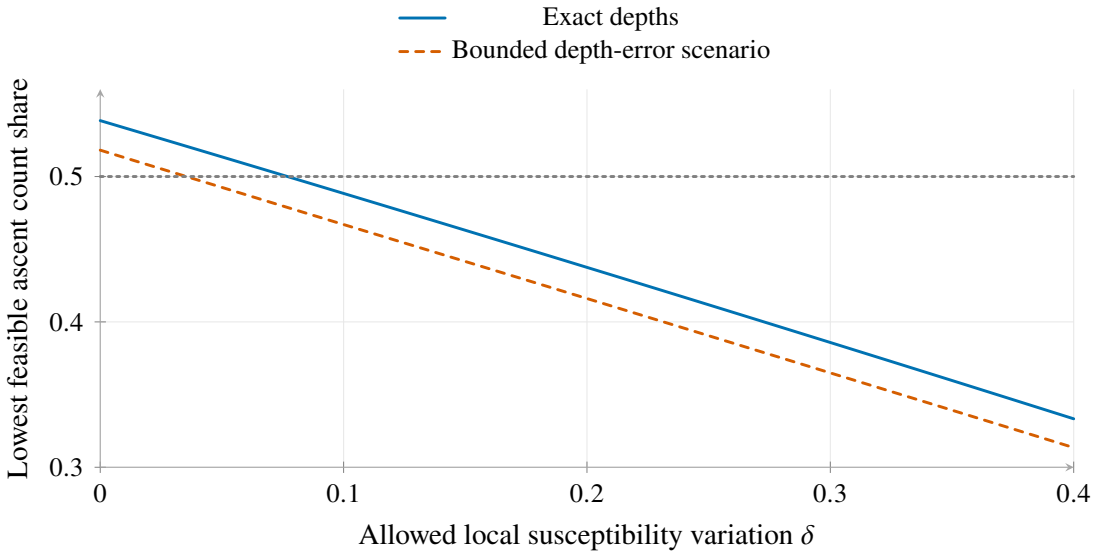


Figure 2. Small allowed material variation can remove an ascent-majority claim.

5 Lateral pathways and simultaneous occupancy

A lateral exposure density $p(x)$ convolved with an individual foot-placement kernel $F(x)$ describes the integrated lateral profile: $L = p * F$. A narrow central pathway often yields a central band; two well-separated pathways yield two bands, as Figure 3 illustrates. The two peaks from alternating feet of one person also broaden or split a single path. Consequently, we compare whole calibrated profiles rather than counting local minima in a noisy surface.

Proposition 5.1 (Concurrency is absent from additive wear)**.** *For fixed contact-loss fields g_j, assigning different occurrence times or grouping contacts into simultaneous events preserves the accumulated field $\sum_j g_j$.*

Proof. The observation is a commutative sum. No event time or pairing variable appears. Regrouping and reordering its summands leaves it unchanged. □

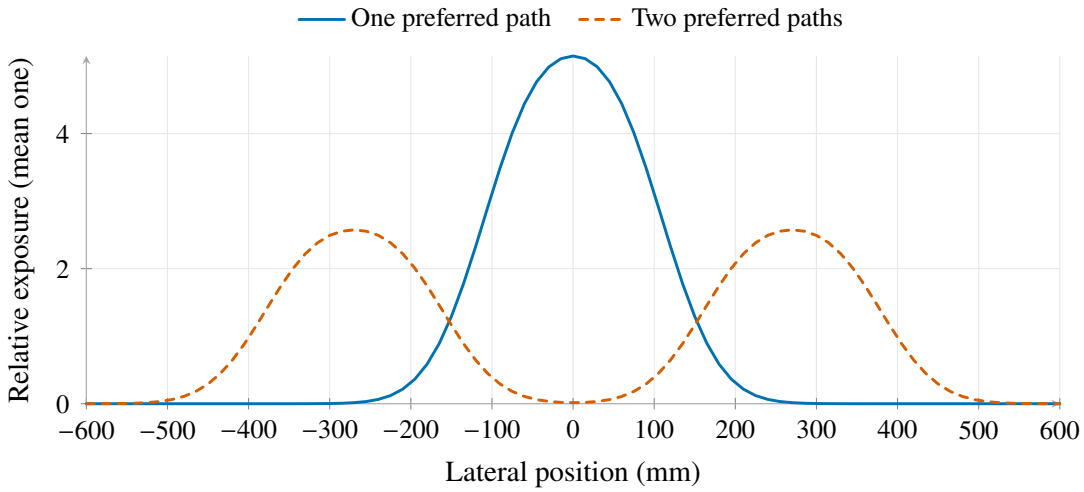


Figure 3. One and two preferred pathways produce different lateral exposure.

For example, 500 side-by-side pairs using left and right pathways and 1,000 sequential single-person traversals alternating those pathways generate identical accumulated loss. The same argument applies to alternating one-way periods versus simultaneous ascent and descent, provided their contact multisets match. More depth points cannot recover temporal associations absent from the observation equation.

The valid answer is therefore "two pathways compatible with side-by-side use," not "two people were simultaneous." Geometry can constrain feasibility. For usable width W_c, side margin m and assumed body envelope b, a conventional side-by-side bound is

$$k_{\max} = \left\lfloor \frac{W_c - 2m}{b} \right\rfloor . \tag{13}$$

For $W_c = 1.20\,\text{m}$, $m = 0.05\,\text{m}$ and stipulated $b = 0.50\,\text{m}$, at most two such envelopes fit. This says that two abreast are feasible under the stated posture and spacing, not that they occurred. It does not bound all people occupying the entire flight in a line. Handrails, carrying loads and turning postures change the assumed envelope.

If simultaneous loading creates an independently calibrated nonlinear wear term, concurrency might enter the model. Yet a dependence of final wear on crowding is additional evidence, not a consequence of lateral lanes. Historical imagery, activity descriptions, width changes and recorded passing practices can resolve some ambiguity. A present-day short observation measures current behavior only. We report it separately from a claim about historic concurrency.

6 An inexpensive, non-destructive survey

6.1 Two people and a minimal field kit

One person records positions and images; another checks the scale, reference patches and repeated profiles. The basic kit is an existing phone camera, measuring tape, rigid straightedge with a mil-

limetre gap scale, a small level, flashlight and 10–20× hand lens. No drilling, scraping, acid tests, hardness indentations or cores are required. Freestanding scale bars and padded straightedge contacts on sound, approved edges avoid adhesive targets and pressure on fragile surfaces. Where contact is unsafe, calibrated side-on photography replaces the gap reading.

First record every tread's width, depth, rise, nosing, slope, joints, orientation, cracks, coatings, drainage and contact restrictions. Photograph the flight, landings and protected sides. Treat different materials and unusual geometries as separate groups. Establish at least three protected, non-collinear reference points where possible, and record why they approximate the original surface. If no such surface survives, retain multiple plausible references and report only relative depressions until absolute loss is bounded.

On each representative tread, obtain a lateral profile near the nosing, at mid-depth and near the rear, plus front-to-back profiles through each worn pathway and a protected region. Use approximately 20–30 mm spacing near depressions and boundaries; coarser points suffice on smooth intact areas. These spacings are starting choices, adjusted until volume and shape contrasts stabilize. Repeat a subset after removing and replacing the instrument. Survey at least one apparently intact tread and both ends of the flight. Continuous photogrammetry is optional: Historic England describes accessible structure-from-motion recording [3], but its accuracy must be checked with held-out scales and independent profile measurements. Pixel count is not metrological accuracy.

The field visit also collects independent historical input: construction and intervention intervals, entrances and exits, public access days, activity type, occupancy ranges and plausible trips per person. Existing quarry records, source specimens and wood identification records can often be consulted without another instrument. Data collection should first target the uncertainty that limits the answer: absolute datum for volume, applicable kernels for direction, q for traffic, and chronology for time.

6.2 Uncertainty, controls and measurement value

For discrete quadrature weights $\mathbf{v}$ in mm^2, $\widehat{V} = \mathbf{v}^T\mathbf{h}$. With depth covariance Σ_h, the measurement variance is $\mathbf{v}^T\Sigma_h\mathbf{v}$. A shared datum error of standard deviation u_b contributes $A^2u_b^2$, so it does not disappear when the grid is refined. This follows standard measurement-uncertainty propagation [4]. A bound $|b| \le \delta_b$ instead gives $|\Delta V| \le A\delta_b$ without requiring a probability model.

For the synthetic area $A = 360{,}000\,\text{mm}^2$, a 0.20 mm reference bias shifts volume by $72\,\text{cm}^3$, or 12% of $600\,\text{cm}^3$. By comparison, adding more independent 0.15 mm readings can reduce random error. The priority is a protected reference and repeated registration, not an arbitrarily dense map. The illustrative 580–620 cm^3 interval in Section 3 would require a tighter reference than the 0.20 mm challenge; it is not derived from that challenge.

Protected patches estimate weathering only when their exposure is comparable. A sheltered side face is not automatically a control for rain on a horizontal tread. Record this difference and vary the environmental correction. Spatially localized residuals suggest a missing mechanism; coherent smooth residuals can indicate reference geometry or weathering. Independent profile measurements

and alternative reference reconstructions should be retained rather than selecting only the scan that best fits the wear model.

We tested the value of sparse sampling on the disclosed synthetic tread. A 24-point design uses six lateral positions $\{-495, -270, -105, 105, 270, 495\}$ mm and four depths $\{60, 97.5, 202.5, 240\}$ mm. It samples both directional bands and their flanks; these exact positions are a design example, not a prescription for every tread. Unweighted least squares estimates the two mean-one kernels and constant/tilt nuisance fields. Table 4 reports 800 noise realizations for each design. Independent errors have standard deviation 0.15 mm. The correlated alternative has covariance $0.15^2 \exp[-(\Delta x/75)^2/2 - (\Delta y/30)^2/2]$ in mm^2, plus independent 0.05 mm noise. A common datum shift of standard deviation 0.20 mm is absorbed by the fitted constant. Under correlated errors,

Table 4. Conditional sampling precision for a true wear share of 0.700.

Design and noise	Points	Share SD	Empirical 2.5–97.5 percentiles
Dense, independent	3,321	0.00058	0.6988–0.7011
Sparse, independent	24	0.00513	0.6894–0.7098
Dense, correlated	3,321	0.00487	0.6905–0.7094
Sparse, correlated	24	0.00595	0.6878–0.7115

the sparse design's standard deviation is only 1.22 times that of the dense scan, despite 138 times fewer readings. Dense-scan uncertainty increases 8.46 times relative to independent noise. A covariance calculation using the least-squares influence vector independently agreed with all four simulated standard deviations within 3%. These results favor well-placed repeatable profiles when dense-scan errors are correlated; they do not guarantee this precision for a hand scale. Instrument resolution, protected datum and actual repeatability must be measured first.

Putting 24 readings along $y = 150$ mm instead makes the two symmetric directional kernels identical there. The projected second singular value is about 10^{-15}, compared with 12.4 for the dispersed design. More precise central readings cannot rescue missing front-to-back information. Neither sparse nor dense sampling resolves the kernel or material counterexamples in Section 4; the value of a reading depends on the ambiguity it can remove.

7 Historical consistency, age and intensity over time

7.1 A consistency test that respects uncertain history

Let $\mathcal{H}$ be the set of histories allowed by documentary evidence, calibration ranges, access patterns and possible interventions. Define

$$\mathcal{F} = \left\{ \eta \in \mathcal{H} : \min_{\mathbf{c}} \|\mathbf{w} - \mathbf{g}(\eta) - B\mathbf{c}\|_{\Sigma^{-1}} \le \tau \right\}, \tag{14}$$

where $\mathbf{g}$ predicts the measured surface and τ is selected from actual uncertainty or a stated tolerance. A nonempty $\mathcal{F}$ establishes compatibility within this model and tolerance. An empty set rejects the

joint assumptions, prompting a check of date, traffic, material, environment or repair. It does not by itself tell which assumption failed. A coarse finite scenario grid cannot prove emptiness over an unsearched continuous set.

In the scalar example, documentary daily traversals of 70–140, age 300–500 years and the stipulated q range predict 153.405–2,045.4 cm^3. This intersects observed 580–620 cm^3, so the volume budget is compatible but weakly restrictive. If all evidence instead constrained $q \leq 0.01\,\mathrm{mm}^3$, daily traversals to at most 100 and exposure to at most 200 years, the greatest possible loss would be 73.05 cm^3. It would conflict with the observed lower bound by 506.95 cm^3. Such a contradiction is useful because it identifies the size of the missing mechanism or incorrect history.

Full spatial compatibility is stronger than volume compatibility. In Table 3, the changed kernel can produce a similar total loss while failing the depth pattern. Conversely, the material counterexample passes even a full-field check. We therefore combine independent history with shape and material observations; no single good-fit statistic replaces them.

7.2 Age estimates and their reliability

For independently bounded daily rate $n \in [n_-, n_+]$ and $q \in [q_-, q_+]$, constant-use exposure age obeys

$$T \in \left[\frac{V_-^{\mathrm{tr}}}{365.25 q_+ n_+}, \frac{V_+^{\mathrm{tr}}}{365.25 q_- n_-} \right]. \tag{15}$$

The illustrative 580–620 cm^3 and 70–140 daily traversals yield approximately 141.8–1,212.5 years under the same q range. Intersecting with the prior documentary interval $[300, 500]$ years leaves that entire interval. Wear provides no chronological tightening in this example. The result should not be reported as a precise "wear age" near 400 years.

The wear exposure age can differ from construction age through delayed opening, closure, reused material and surface renewal. A replacement tread dates the retained surface history; it does not reset the whole building. Uniform resurfacing can remove the earlier profile and its original reference, erasing chronology even if a modern surface looks regular. Calibration uncertainty dominates reliability when it is broad, and shared error across treads prevents twelve treads from acting as twelve independent clocks.

Proposition 7.1 (Time compression)**.** *With fixed kernels and coefficients, traffic $n_d(t)$ on $[0, T]$ and compressed traffic $n'_d(t) = c\, n_d(ct)$ on $[0, T/c]$ produce identical traffic wear for any $c > 0$.*

Proof. Changing variables $s = ct$ gives $\int_0^{T/c} c\, n_d(ct)dt = \int_0^T n_d(s)ds$. Each directional contribution in Equation (2) is unchanged. □

Consequently, 15 million traversals spread over 400 years at 102.67 per day and over 40 years at 1,026.69 per day have the same abrasion field. Likewise, an annual festival can match evenly distributed use with the same contacts. Known closures, phased construction, changes in footwear, weathering exposure or dated intervention surfaces can constrain timing, but only when these additional mechanisms are independently measured. Weathering could act as a second clock if its rate

and controls were calibrated; allowing an arbitrary weathering rate merely adds another unknown product.

Nonlinearity alone does not create a calendar clock. For an autonomous exposure law $\dot{h} = n(t)F(h)$ with a unique solution, let cumulative exposure be $\tau(t) = \int_0^t n(s)ds$. The chain rule gives $dh/d\tau = F(h)$, so the final field depends on total exposure rather than its temporal compression. The saturation challenge in Table 3 is of this form. A mechanism that changes with elapsed calendar time, traffic rate or order, together with its calibration, is needed to distinguish concentrated from sustained use. This qualification avoids treating every failure of linear abrasion as evidence of traffic timing.

8 Repairs and renovations: locate anomalies, then distinguish causes

Fit a common normalized traffic field within a group of geometrically comparable treads, allowing an individual amplitude. Compare each amplitude with the group's median and examine shape residuals. A low-amplitude tread or an abrupt change in material, nosing geometry, joint lines and tool marks is a repair candidate. A patch boundary that cuts across a continuous worn pathway is stronger evidence than smooth amplitude variation alone.

The synthetic twelve-tread test varied intact amplitudes by about 4% and set tread six to half retained loss. Figure 4 shows the recovered median-normalized amplitude of 0.5004 on that tread; other values lay between 0.9599 and 1.0398. The anomaly is clear relative to the specified noise and variability. Yet a twice-as-resistant tread, a history with half the contacts, and a replacement retaining half the exposure all give the same scaled field. The experiment detects a candidate; it does not identify the cause from amplitude alone.

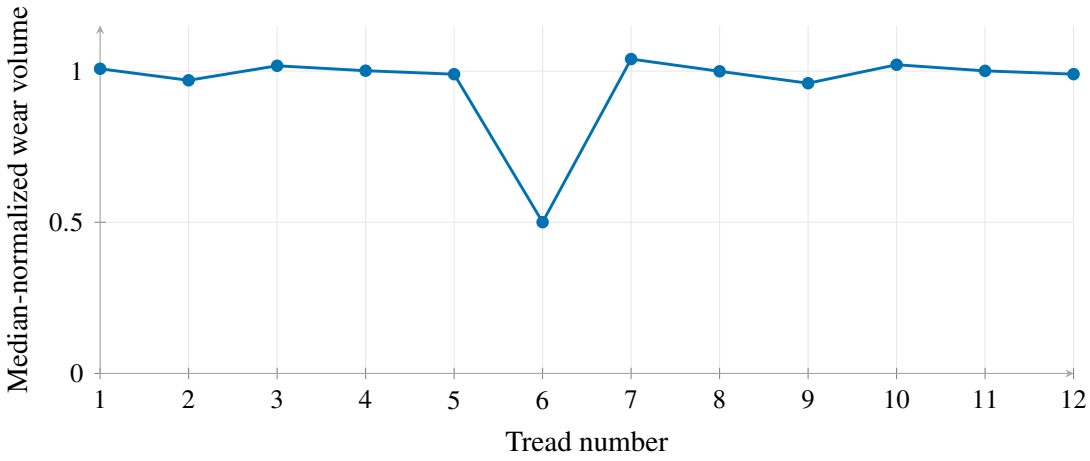


Figure 4. A low-wear tread is an intervention candidate, not a unique diagnosis.

For epochs j, a retained-volume budget is

$$V_i^{\mathrm{tr,ret}} = \sum_j r_{ij} \sum_d q_{ij,d} N_{j,d}, \quad 0 \le r_{ij} \le 1. \tag{16}$$

A full replacement sets earlier retention to zero; a partial repair affects only its patch. The scalar r_{ij} approximates proportional retention; actual patch changes require spatial masks. This prevents old and new contact histories from being forced into a single false date. With known time since replacement and shared post-repair traffic, V_i/q_i estimates post-repair exposure relative to intact treads. Without intervention dates or q_i, the reconstruction remains a family of histories.

We recommend an evidence hierarchy. Consistent joints, a distinct insert, independent records and a material boundary support an intervention. An amplitude or residual anomaly supplies supporting evidence. A low fit error after introducing a repair parameter is weakest because that parameter can absorb hardness, avoidance or reference error. A report should map suspected interventions, state their competing explanations and preserve the original measurements.

9 Material provenance and wood-specific chronology

9.1 Candidate stone sources

Wear can test whether a claimed source has compatible abrasion behavior under a specified contact and exposure history. It cannot name a quarry from hardness or volume alone. If two candidate quarries admit coefficients q_1 and q_2, Equation (4) lets different traffic counts reproduce the same loss. Surface weathering can also alter material appearance and susceptibility.

The basic survey records visible mineral assemblage, grain size, bedding, inclusions, fossils where present, color on protected patches and orientation. Compare these with existing geological or quarry specimens and archival building records. Do not break a fresh surface to improve visibility. A candidate with an incompatible visible assemblage can be excluded; several visually compatible sources should remain listed. Wear compatibility is then the feasible set of source-specific coefficients, chronological ranges and traffic bounds in Equation (14).

If source identity is critical and equipment can be borrowed, non-invasive compositional analysis can add evidence. X-ray fluorescence provides elemental features for comparison with source materials [5]. Reference-source variability, instrument geometry and weathered surfaces must be included; a matching fingerprint supports compatibility, especially when alternative sources overlap. This optional equipment is not part of the minimal low-cost protocol. Thin-section petrography requires a physical sample [6], so it is excluded on the stair itself; an existing archived section may be consulted.

9.2 Wood species, tree age and installation date

For wood, visible end grain and existing exposed faces can reveal vessel, ray and growth-ring structures with a hand lens. The USDA Wood Handbook describes hand-lens identification and cautions

that routine anatomical identification is often reliable at genus rather than species level [7]. Surface color and nominal hardness alone are inadequate. Grain direction, moisture, coating, local rot and prior sanding can change loss; coefficients from fresh laboratory wood do not automatically apply to an aged tread.

Photograph an already exposed ring sequence without scraping or coring. If sufficiently clear and long, a specialist can compare widths with appropriate reference chronologies. Tree-ring dating is an independent chronology method [8]. A preserved outermost ring or bark edge can constrain felling; with missing outer rings, the last preserved dated ring gives a lower bound on the felling year, and missing pith prevents a complete tree-age count. Felling precedes installation, possibly by a long interval if timber was stored or reused. Counting visible rings is therefore neither a construction date nor necessarily the age of the tree.

The archaeological conclusion should combine anatomical compatibility, any accessible independent chronological match, joinery and repair records with wear bounds. The inexpensive protocol may stop at "compatible with this genus and these candidate histories." It must not force an exact species, tree age or source location from a surface that does not contain those distinctions.

10 Validation, scope and final guidance

Our evidence contains analytic nonidentifiability results, synthetic structural challenges and independent numerical reconstruction. There is no measured historical stair dataset to validate an ancient traffic estimate. The strongest unconditional conclusions are limits and an observation design; directional and chronological estimates remain conditional on field evidence.

The independent audit reconstructs coefficients using normal equations, volume integrals using Simpson quadrature, scalar intervals by corner enumeration and projected singular values with a Schur complement. It checks contact-multiset equality, bounded-material intervals without using the author's linear program, stored primal/dual certificates and analytic survey covariance against simulation. Changed and cross-mixed kernels, exposure saturation, material alternatives and repair amplitude ambiguity address different failure mechanisms. The separate coordinating assistant also reproduced the material intervals by amplitude feasibility and bisection; this is computational review of shared synthetic kernels, not human or field validation.

For an archaeologist, the answer is organized by what the stairs remember. Loss volume can become a traversal interval after contact calibration, and then a daily-rate interval after chronological evidence. Direction can become a feasible count-share range when kernels survive the nuisance-rank and material checks. Lateral lanes describe preferred paths; they do not determine simultaneous occupancy. Cross-tread anomalies identify places to inspect for repairs, while joints and material contrasts determine whether a repair interpretation is persuasive. Quarry and wood hypotheses require independent material comparison, and the date of a retained surface must be separated from the construction date.

The illustrative 580–620 cm^3 budget admits approximately 40–283 traversals per day under the stated calibration and age ranges. It cannot choose between short intense use and long light

use. A narrow noise interval is not a historical certainty: approximately ±7.69% permitted local susceptibility variation, together with the stipulated reference-coefficient ratio, permits balanced counts in the synthetic example. The sparse-survey result also shows why spatial placement and correlated error matter more than pixel count. Our next observations constrain original reference geometry, material variation, kernel transfer and documented exposure according to the claim being tested. Additional algorithms cannot replace those missing measurements.

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Report on Use of AI

This report records an autonomous development research exercise using the original 2025 MCM A statement. Codex, an OpenAI GPT-6 based assistant, produced the model formulation, analytic arguments, Python implementation, independent implementation checks, literature searches, manuscript and typesetting review. The user supplied the research scope and constraints; the coordinating assistant assigned the manuscript identifier. No human numerical solving or verification is claimed. The host did not expose a more specific model identifier or metered bill to the execution author.

The operative instruction requested a complete English solution, a valid baseline preserved before revision, bounded structural challenges, a non-destructive low-cost measurement protocol, truthful source exposure and a reproducible final artifact. The assistant used general literature on abrasion, plantar loading, measurement uncertainty, wood identification and archaeological material comparison. No same-question solution was supplied or accessed in this run; the project had historic contact with this question, so this is not described as an unseen evaluation.

All numerical surfaces were generated by the disclosed synthetic mechanisms. The implementation audit was written by the execution assistant using separate formulas; a coordinating assistant independently reconstructed the bounded-material intervals from the saved synthetic fields. Neither audit is an independent human peer review or field validation. Python execution verified specified numerical claims. LaTeX compilation and actual page-by-page visual inspection checked the final artifact. These steps do not validate archaeological assumptions in the absence of measured data.

The retained local source files and execution records document commands, parameters, source ranges, failures and artifact hashes. They are not a complete verbatim transcript of all model interaction. No content was submitted to a contest or published. The appended report is outside the original statement's 25-page solution limit.